

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
df = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/SoyaRe
df.head()
```

	web_scraper_order	web_scraper_start_url	Name	Title
0	1771222379-1	https://www.amazon.in/product-reviews/B09GH5H2...	karen	Great Meat substitute
1	1771222379-2	https://www.amazon.in/product-reviews/B09GH5H2...	Rupali	Taste like soya only
2	1771222379-3	https://www.amazon.in/product-reviews/B09GH5H2...	Technocrat	This keema is tasty and so convenient.
3	1771222379-4	https://www.amazon.in/product-reviews/B09GH5H2...	Leona	The flavours are melting on my tongue!
4	1771222379-5	https://www.amazon.in/product-reviews/B09GH5H2...	Amarnath Banerjee	Too Many Spices !

```
!pip install vaderSentiment
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
```

Collecting vaderSentiment

Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl.metadata (57
Requirement already satisfied: requests in /usr/local/lib/python3.12
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/loca
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/pythor
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/
Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl (125 kB)

126.0/126.0 kB 3.7 MB/s

Installing collected packages: vaderSentiment
Successfully installed vaderSentiment-3.3.2

```
from transformers import pipeline
sentiment_pipeline = pipeline("sentiment-analysis")
```

No model was supplied, defaulted to distilbert/distilbert-base-uncas
Using a pipeline without specifying a model name and revision in pro
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.
The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your se
You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to
warnings.warn(
config.json: 100%

629/629 [00:00<00:00, 53.3kB/s]

model.safetensors: 100%

268M/268M [00:02<00:00, 151MB/s]

Loading weights: 100% 104/104 [00:00<00:00, 163.36it/s, Materializing param=pre_classifier.weig

tokenizer_config.json: 100%

48.0/48.0 [00:00<00:00, 787B/s]

vocab.txt: 232k/? [00:00<00:00, 8.11MB/s]

```
custom_lexicon = {
    "rubbery": -2,
    "meaty": +2,
    "aftertaste": -1,
    "bland": -2,
    "juicy": +2,
    "realistic": +3,
    "chemical": -3
}
```

```
analyzer = SentimentIntensityAnalyzer()
analyzer.lexicon.update(custom_lexicon)
```

```
from sklearn.feature_extraction.text import CountVectorizer
```

```
import spacy  
nlp = spacy.load("en_core_web_sm")
```

```
from sklearn.decomposition import LatentDirichletAllocation
```

```
!pip install bertopic  
from bertopic import BERTopic
```

[Show hidden output](#)

```
def aspect_polarity_vector(text):  
    scores = []  
    text_lower = str(text).lower()  
  
    for aspect, keywords in aspects.items():  
        if any(k in text_lower for k in keywords):  
            s = analyzer.polarity_scores(text_lower)  
            scores.extend([s['pos'], s['neg']])  
        else:  
            scores.extend([0, 0]) # no mention  
  
    return scores  
  
df['aspect_vector'] = df['Review'].apply(aspect_polarity_vector)
```

```
aspects = {  
    "taste": ["taste", "flavour", "flavor", "bland", "yummy"],  
    "texture": ["texture", "rubbery", "chewy", "soft", "hard"],  
    "smell": ["smell", "odor", "odour", "aroma"],  
    "price": ["price", "cost", "expensive", "cheap"],  
    "health": ["healthy", "protein", "processed", "nutrition"]  
}
```

```
def aspect_polarity_vector(text):
    scores = []
    text_lower = str(text).lower()

    for aspect, keywords in aspects.items():
        if any(k in text_lower for k in keywords):
            s = analyzer.polarity_scores(text_lower)
            scores.extend([s['pos'], s['neg']])
        else:
            scores.extend([0, 0]) # no mention

    return scores

df['aspect_vector'] = df['Review'].apply(aspect_polarity_vector)
```

```
!pip install pandas vaderSentiment
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/c
Requirement already satisfied: vaderSentiment in /usr/local/lib/pyth
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/pythc
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/pythor
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/pyth
Requirement already satisfied: requests in /usr/local/lib/python3.12
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/loca
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/pythor
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/
```

```
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyze

# Load your CSV
df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/SoyaReview

# Check columns
print(df.columns)

# Initialize VADER
analyzer = SentimentIntensityAnalyzer()

# Function to compute polarity vector
def get_polarity_vector(text):
    scores = analyzer.polarity_scores(str(text))
    return scores # returns dict with pos, neg, neu, compound

# Apply to your review column (change 'review_text' if needed)
df['polarity'] = df['Review'].apply(get_polarity_vector)

# Expand polarity dict into separate columns
```

```
polarity_df = pd.json_normalize(df['polarity'])
```

```
# Merge back to main dataframe
```

```
df = pd.concat([df, polarity_df], axis=1)
```

```
# Show output
```

```
df[['Review', 'pos', 'neg', 'neu', 'compound']].head(10)
```

```
Index(['web_scraper_order', 'web_scraper_start_url', 'Name', 'Title',  
      'Star', 'Verified', 'Date'],  
      dtype='object')
```

	Review	pos	neg	neu	compound
0	Really loved the kit . Easy to use and tastes...	0.268	0.000	0.732	0.9585
1	Saw many reviews of YouTube and decided to giv...	0.056	0.090	0.855	-0.4965
2	We loved this product to bits (pun intended)! ...	0.183	0.029	0.788	0.9358
3	Tastes veryyyyy similar to meat but no cruelty...	0.361	0.000	0.639	0.6716
4	The spices included with the pack must cost mo...	0.126	0.000	0.874	0.7345
5	This is just soyabean same as other soyabean c...	0.065	0.104	0.831	-0.7059
6	Really good texture and excellent quality, val...	0.508	0.000	0.492	0.8516
7	The product came in a good box with all the in...	0.061	0.098	0.841	-0.6859
8	NaN	0.000	0.000	1.000	0.0000
9	Damn I liked it!! It's quite tasty. One star l...	0.181	0.032	0.786	0.8185

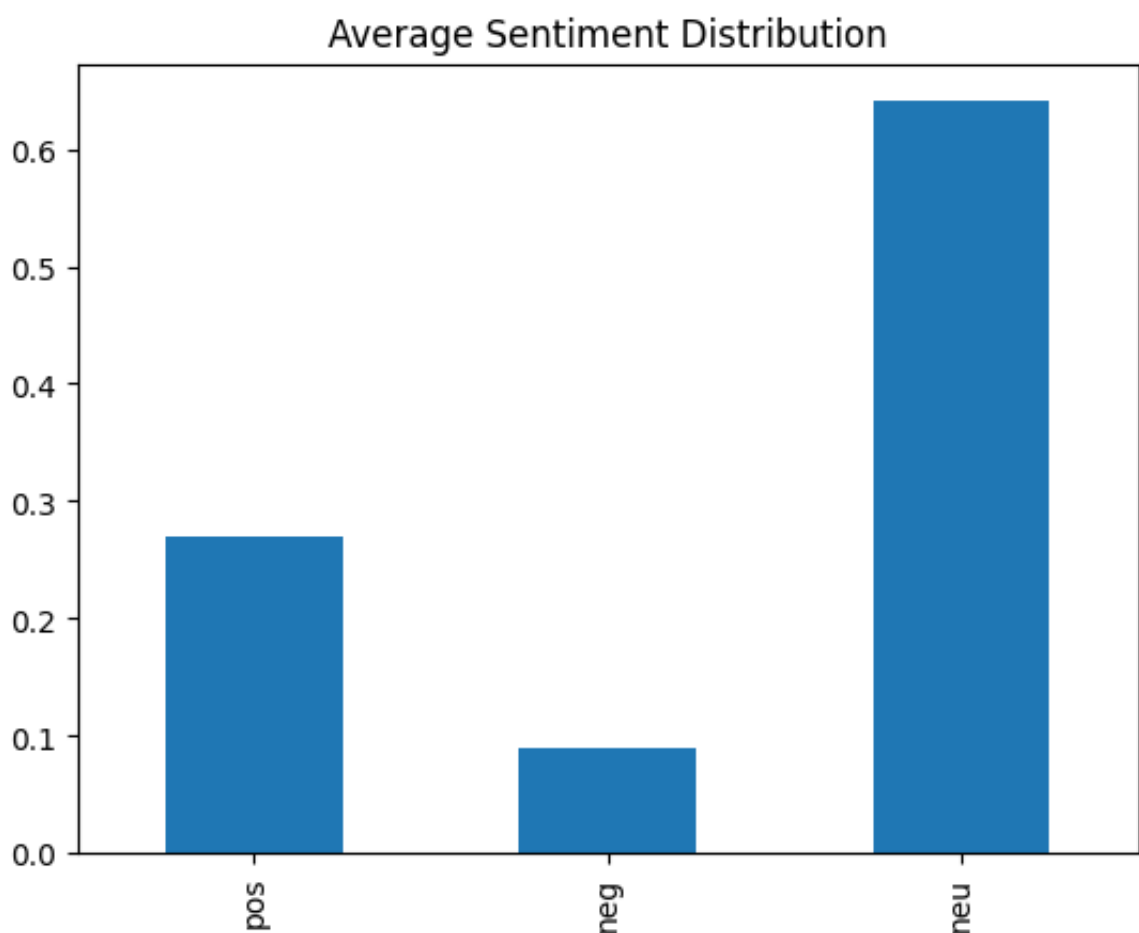
```
df['polarity_vector'] = df.apply(
    lambda row: [row['pos'], row['neg'], row['neu'], row['compound']]
)

df[['Review', 'polarity_vector']].head(5)
```

	Review	polarity_vector
0	Really loved the kit . Easy to use and tastes...	[0.268, 0.0, 0.732, 0.9585]
1	Saw many reviews of YouTube and decided to giv...	[0.056, 0.09, 0.855, -0.4965]
2	We loved this product to bits (pun intended)! ...	[0.183, 0.029, 0.788, 0.9358]
3	Tastes veryyyyyy similar to meat but no cruelty...	[0.361, 0.0, 0.639, 0.6716]
4	The spices included with the pack must cost mo...	[0.126, 0.0, 0.874, 0.7345]

```
import matplotlib.pyplot as plt

df[['pos', 'neg', 'neu']].mean().plot(kind='bar', title='Average Sent
plt.show()
```



```
def label_sentiment(compound):
    if compound >= 0.05:
        return "Positive"
    elif compound <= -0.05:
        return "Negative"
    else:
        return "Neutral"

df['sentiment_label'] = df['compound'].apply(label_sentiment)

df[['Review', 'compound', 'sentiment_label']].head(10)
```

	Review	compound	sentiment_label
0	Really loved the kit . Easy to use and tastes...	0.9585	Positive
1	Saw many reviews of YouTube and decided to giv...	-0.4965	Negative
2	We loved this product to bits (pun intended)! ...	0.9358	Positive
3	Tastes veryyyyy similar to meat but no cruelty...	0.6716	Positive
4	The spices included with the pack must cost mo...	0.7345	Positive
5	This is just soyabean same as other soyabean c...	-0.7059	Negative
6	Really good texture and excellent quality, val...	0.8516	Positive
7	The product came in a good box with all the in...	-0.6859	Negative
8	NaN	0.0000	Neutral
9	Damn I liked it!! It's quite tasty. One star I...	0.8185	Positive